

# Synopsis

## Project Title

### Student Management System using Python (Console Based)

Prepared By: Avantika Yadav

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#### Introduction

The Student Management System is a console-based application developed using Python. It helps manage student information such as student details, marks, and results through different administrator roles. The project is developed using Object-Oriented Programming (OOP) concepts without using a database.

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#### Problem Statement

Managing student records manually can be difficult and time-consuming. This project provides a simple and organized way to maintain student information using Python.

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#### Objectives

- To manage student records efficiently.
  - To provide different access levels for administrators.
  - To perform student record operations easily.
  - To practice core Python concepts.
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#### Scope

The project can be used by schools or colleges to manage student records. It is suitable for learning Python programming concepts and can be enhanced in the future by adding a database and GUI.

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#### Technologies Used

- Python 3
- OOP (Classes & Objects)
- Functions

- Loops
  - Conditional Statements
  - Exception Handling
  - Lists
  - Dictionaries
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## **Admin Roles**

### **1. Super Admin**

- Login
  - Add Student
  - Delete Student
  - Update Student
  - Search Student
  - View All Students
  - Manage Admins
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### **2. HOD**

- Login
  - Add Student
  - Update Student
  - Search Student
  - View Student Details
  - View Result
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### **3. Teacher**

- Login
- Add Marks
- Update Marks
- View Student

- Display Result
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### **Student Details**

- Student ID
  - Name
  - Age
  - Gender
  - Class
  - Section
  - Roll Number
  - Marks
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### **Main Modules**

- Admin Login
  - Student Registration
  - Student Search
  - Update Student
  - Delete Student
  - Display Student Details
  - Marks Management
  - Result Calculation
  - Logout
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### **Python Concepts Used**

- Classes & Objects
- Constructors
- Attributes
- Methods
- Functions

- Loops
  - Conditional Statements
  - Lists
  - Dictionaries
  - Exception Handling
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### **Expected Outcome**

The system will provide a simple interface for managing student information through different administrator roles. It will demonstrate the practical implementation of Python OOP concepts without using a database.

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### **Future Enhancements**

- File Handling
  - MySQL Database
  - Tkinter GUI
  - Student Login
  - Attendance Management
  - Report Card Generation
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### **Conclusion**

The Student Management System is a beginner-friendly Python project that demonstrates the implementation of OOP concepts through a real-world application. It helps in understanding classes, objects, methods, and Python programming while providing an organized way to manage student records.